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My research focuses on **steered generative modeling** and **graph-structured learning**, with applications to **neural architecture search** and **anomaly detection** on **dynamic networks**:

- RL-steered **graph diffusion** for **neural architecture search** on directed acyclic graphs (DGPO - matches NAS-Bench-201 optima; transferable structural priors from 7% of search space). *Accepted, IJCNN 2026.*
- Unsupervised **anomaly detection on temporal graphs**: TGN-SVDD and its probabilistic extension RTGN-SVDD for **robust intrusion detection** under synthetic noise injection and adversarial perturbation.
- Multi-objective **AutoML** pipelines for Pareto-optimal **accuracy-sparsity trade-offs** for interpretable sparse models, evaluated on open-source cybersecurity datasets (TPOT, Optuna, NSGA-II).

Experience

Nov 2020 - Present **PhD Candidate - Machine Learning**, *Bielefeld University*, Bielefeld, Germany
- Research on generative AI for neural architecture search (graph diffusion + RL), anomaly & OOD detection with AutoML, GNNs, and probabilistic methods.
- 6 published papers (5 first-authored). Supervised 1 BSc and 1 MSc thesis.
- Designed and executed large-scale training pipelines on multi-GPU SLURM-managed clusters.
- Built AI-augmented research workflows (Claude Code, Codex) to automate experimental pipelines, from implementation through systematic evaluation.

Oct 2023 - Sep 2025 **Tutor & Supervisor**, *Bielefeld University*, Bielefeld, Germany
Applied Optimization (Oct 2023 - Mar 2025): designed exercises, supervised tutorials. AI Act in ML Seminar (Apr 2024 - Sep 2025): led seminar on EU AI regulation; supervised student projects.

Sep 2011 - Jul 2018 **Founder & CEO - Baltik Bread**, *Bakery Manufacturing & Mobile Retail*, Novosibirsk, Russia
- Co-founded and scaled a dual manufacturing + retail operation from zero; 500k-600k EUR investment; full ROI within 4 years.
- Grew to ~1M EUR annual revenue, ~2,500 units/day across ~40 SKUs with 90%+ sell-through; built and managed a 25-person team.
- Data-driven location optimization using empirical sales analysis; built competitive advantage through systematic testing.

Education

Nov 2020 - Present **PhD (ongoing) in Machine Learning**, *Bielefeld University*, Bielefeld, Germany, Expected Summer 2026
Machine Learning Group (HammerLab), supervised by Prof. Barbara Hammer.

Oct 2018 - Nov 2020 **M.Sc. Data Science**, *Bielefeld University*, Bielefeld, Germany
Thesis: Automated Machine Learning (AutoML) technologies for feature selection.

Sep 2007 - Jun 2010 **M.Sc. Econometrics & Quantitative Economics**, *Novosibirsk State University*, Novosibirsk, Russia

Sep 2003 - Jun 2007 **B.Sc. Physics**, *Novosibirsk State University*, Novosibirsk, Russia

Publications

- **A. Liuliakov**, L. Hermes, and B. Hammer. "DGPO: RL-Steered Graph Diffusion for Neural Architecture Generation." *IJCNN 2026.*
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. "Noise Robust One-Class Intrusion Detection on Dynamic Graphs." In: *ESANN 2024.*

- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. "One-Class Intrusion Detection with Dynamic Graphs." In: *ICANN 2023*. [code]
- L. Hermes, **A. Liuliakov**, and M. Schilling. "Graph Learning by Dynamic Sampling." In: *IJCNN 2023*.
- **A. Liuliakov**, L. Hermes, and B. Hammer. "AutoML Technologies for the Identification of Sparse Classification and Outlier Detection Models." In: *Applied Soft Computing*, 2023.
- **A. Liuliakov** and B. Hammer. "AutoML Technologies for the Identification of Sparse Models." In: *IDEAL 2021*.

Peer reviewing: ESANN 2024 (2 papers), IJCNN 2024 (1), ESANN 2026 (2), IJCNN 2026 (1).

Skills

ML & AI	Generative AI, Diffusion Models, Reinforcement Learning, Graph Neural Networks, Deep Learning, AutoML, Neural Architecture Search, Anomaly Detection, Probabilistic Modeling
Programming	Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Optuna, scikit-learn, NumPy, Pandas, Linux, Slurm, Git, W&B
AI Tools	Claude Code, OpenAI Codex, GitHub Copilot; custom agentic workflows for automated experimental pipelines and cluster deployment

Languages

English	Full professional proficiency (C1)
German	Limited working proficiency (B1)
Russian	Native